

NEHA SHARMA - U24AI050 LAB-4

16 February 2026 11:53

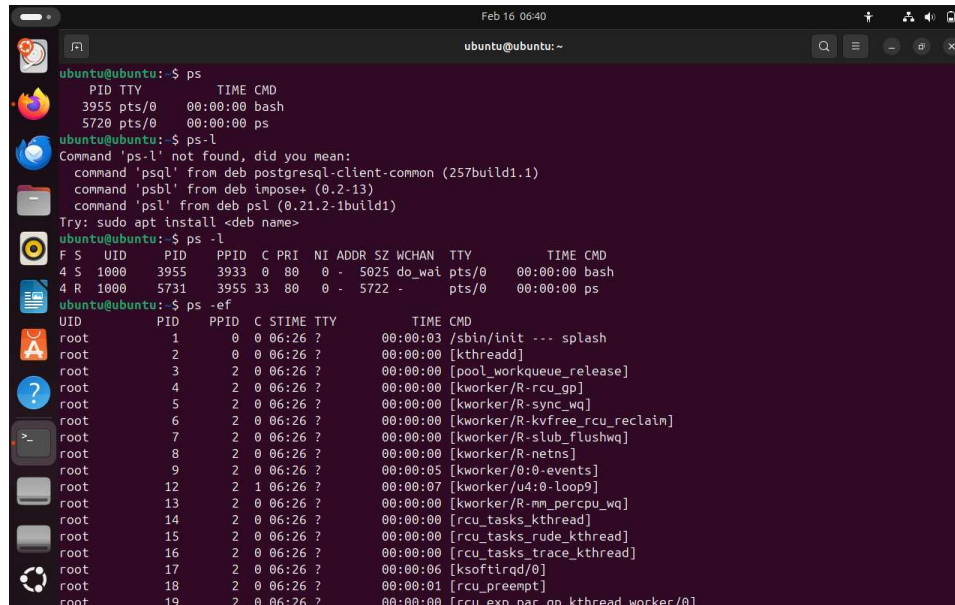
1. Perform Following

a) Obtain a listing of processes by using the ps command.

ps

ps -l

ps -ef

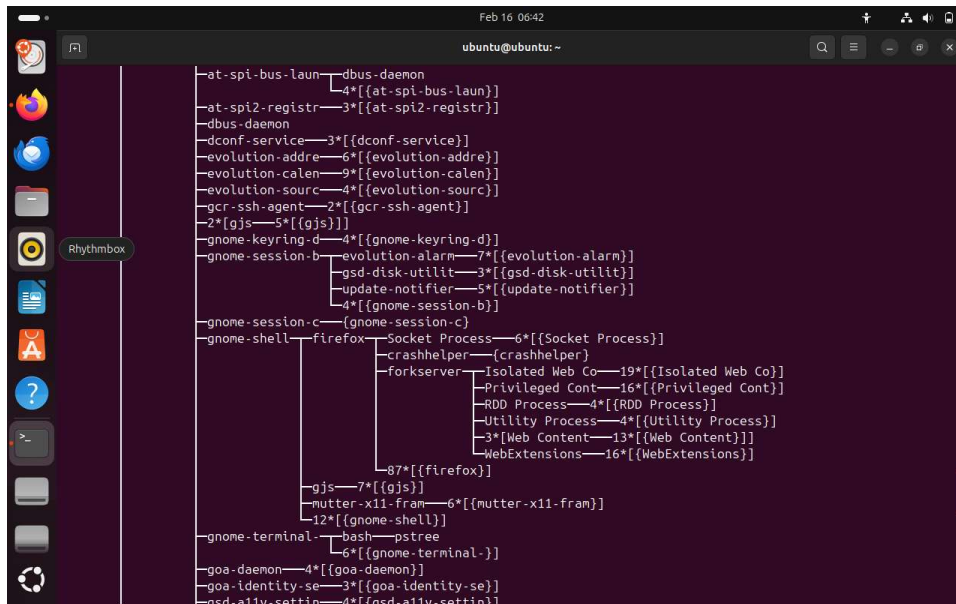


```
ubuntu@ubuntu:~$ ps
  PID TTY          TIME CMD
 3955 pts/0    00:00:00 bash
 5720 pts/0    00:00:00 ps
ubuntu@ubuntu:~$ ps -l
Command 'ps-l' not found, did you mean:
  command 'psql' from deb postgresql-client-common (257build1.1)
  command 'psbl' from deb inps+ (0.2-13)
  command 'psl' from deb psl (0.21.2-1build1)
Try: sudo apt install <deb name>
ubuntu@ubuntu:~$ ps -l
 F S   UID        PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
 4 S   1000        3955    3933  0  80   0  - 5025 do_wai pts/0    00:00:00 bash
 4 R   1000        5731    3955  33  80   0  - 5722 -      pts/0    00:00:00 ps
ubuntu@ubuntu:~$ ps -ef
UID        PID  PPID  C STIME TTY          TIME CMD
root         1      0  0 06:26 ?        00:00:03 /sbin/init --- splash
root         2      0  0 06:26 ?        00:00:00 [kthreadd]
root         3      2  0 06:26 ?        00:00:00 [pool_workqueue_release]
root         4      2  0 06:26 ?        00:00:00 [kworker/R-rcu_gp]
root         5      2  0 06:26 ?        00:00:00 [kworker/R-sync_wq]
root         6      2  0 06:26 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root         7      2  0 06:26 ?        00:00:00 [kworker/R-slub_flushwq]
root         8      2  0 06:26 ?        00:00:00 [kworker/R-netns]
root         9      2  0 06:26 ?        00:00:05 [kworker/0:0-events]
root        12      2  1 06:26 ?        00:00:07 [kworker/u4:0-loop9]
root        13      2  0 06:26 ?        00:00:00 [kworker/R-mm_percpu_wq]
root        14      2  0 06:26 ?        00:00:00 [rcu_tasks_kthread]
root        15      2  0 06:26 ?        00:00:00 [rcu_tasks_rude_kthread]
root        16      2  0 06:26 ?        00:00:00 [rcu_tasks_trace_kthread]
root        17      2  0 06:26 ?        00:00:06 [ksoftirqd/0]
root        18      2  0 06:26 ?        00:00:01 [rcu_preempt]
root        19      2  0 06:26 ?        00:00:00 [rcu_exp_nar_gp_kthread worker/0]
```

```
Feb 16 06:41
ubuntu@ubuntu:~
systemd+ 1307 1 0 06:27 ? 00:00:00 /usr/lib/systemd/systemd-resolved
systemd+ 1308 1 0 06:27 ? 00:00:00 /usr/lib/systemd/systemd-timesyncd
avahi 1551 1 0 06:27 ? 00:00:00 avahi-daemon: running [ubuntu.local]
message+ 1552 1 0 06:27 ? 00:00:01 @dbus-daemon --system --address=systemd: --nofork --nopidfile --syst
gnome-r+ 1555 1 0 06:27 ? 00:00:00 /usr/libexec/gnome-remote-desktop-daemon --system
polkitd 1557 1 0 06:27 ? 00:00:00 /usr/lib/polkit-1/polkitd --no-debug
root 1558 1 0 06:27 ? 00:00:00 /usr/libexec/power-profiles-daemon
root 1562 1 0 06:27 ? 00:00:00 /usr/libexec/accounts-daemon
root 1566 1 0 06:27 ? 00:00:00 /usr/sbin/cron -f -P
root 1569 1 0 06:27 ? 00:00:00 /usr/libexec/switcheroo-control
root 1576 1 0 06:27 ? 00:00:00 /usr/lib/systemd/systemd-logind
root 1585 1 0 06:27 ? 00:00:00 zed -F
avahi 1595 1551 0 06:27 ? 00:00:00 avahi-daemon: chroot helper
root 1596 1 0 06:27 ? 00:00:00 /usr/sbin/NetworkManager --no-daemon
root 1602 1 0 06:27 ? 00:00:00 /usr/sbin/wpa_supplicant -u -s -O DIR=/run/wpa_supplicant GROUP=netd
syslog 1655 1 0 06:27 ? 00:00:00 /usr/sbin/rsyslogd -n -iNONE
root 1664 1 0 06:27 ? 00:00:00 /usr/sbin/ModemManager
root 1672 1 0 06:27 ? 00:00:00 /usr/libexec/udisks2/udisksd
root 1833 1 0 06:28 ? 00:00:00 /usr/sbin/cupsd -l
root 1835 1 0 06:28 ? 00:00:01 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s
cups-br+ 2174 1 0 06:28 ? 00:00:00 /usr/sbin/cups-browsed
kernoops 2195 1 0 06:28 ? 00:00:00 /usr/sbin/kerneloops --test
kernoops 2203 1 0 06:28 ? 00:00:00 /usr/sbin/kerneloops
root 2265 1 0 06:28 ? 00:00:00 /usr/sbin/gdm3
root 2273 2265 0 06:28 ? 00:00:00 gdm-session-worker [pam/gdm-autologin]
ubuntu 2281 1 0 06:28 ? 00:00:02 /usr/lib/systemd/systemd --user
ubuntu 2284 2281 0 06:28 ? 00:00:00 (sd-pam)
ubuntu 2303 2281 0 06:28 ? 00:00:00 /usr/bin/pipewire
ubuntu 2304 2281 0 06:28 ? 00:00:00 /usr/bin/pipewire -c filter-chain.conf
ubuntu 2306 2281 0 06:28 ? 00:00:00 /usr/bin/wireplumber
ubuntu 2307 2281 0 06:28 ? 00:00:00 /usr/bin/pipewire-pulse
ubuntu 2308 2281 0 06:28 ? 00:00:00 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secre
ubuntu 2313 2281 0 06:28 ? 00:00:01 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfi
```

b) Use a Linux command which displays a tree of all processes in the system.
pstree

```
Feb 16 06:42
ubuntu@ubuntu:~
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ d
d: command not found
ubuntu@ubuntu:~$ pstree
systemd--ModemManager--3*[{ModemManager}]
--NetworkManager--3*[{NetworkManager}]
--accounts-daemon--3*[{accounts-daemon}]
--avahi-daemon--avahi-daemon
--colord--3*[{colord}]
--cron
--cups-browsed--3*[{cups-browsed}]
--cupsd
--dbus-daemon
--gdm3--gdm-session-wor--gdm-x-session--Xorg--{Xorg}
--gnome-session-b--3*[{gnome-session-b}]
--3*[{gdm-x-session}]
--3*[{gdm-session-wor}]
--3*[{gdm3}]
--gnome-remote-de--3*[{gnome-remote-de}]
--2*[{kerneloops}]
--polkitd--3*[{polkitd}]
--power-profiles--3*[{power-profiles-}]
--rsyslogd--3*[{rsyslogd}]
--rtkit-daemon--2*[{rtkit-daemon}]
--snapd--9*[{snapd}]
--subiquity-serve--python3.10--3*[{python3.10}]
--switcheroo-cont--3*[{switcheroo-cont}]
--systemd--(sd-pam)
--at-spi-bus-laun--dbus-daemon
--4*[{at-spi-bus-laun}]
--at-spi2-registr--3*[{at-spi2-registr}]
--dbus-daemon
--dconf-service--3*[{dconf-service}]
```



c) Use Exec(), Wait() and Exit() system call.

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

```

```

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child Process\n");
        execl("/bin/ls", "ls", NULL);
        exit(0);
    }
    else {
        wait(NULL);
        printf("Parent Process\n");
    }

    return 0;
}

```

```
Feb 16 07:03
ubuntu@ubuntu:~
├─upower─3*[{upower}]
├─wpa_supplicant
└─zed─2*[{zed}]
ubuntu@ubuntu:~$ nano exec_example.c
ubuntu@ubuntu:~$ gcc exec_example.c
Thunderbird Mail not found, but can be installed with:
sudo apt install gcc
ubuntu@ubuntu:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
binutils binutils-common binutils-x86-64-linux-gnu dpkg-dev fakeroot g++ g++-13 g++-13-x86-64-linux-gnu
g++-x86-64-linux-gnu gcc gcc-13 gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libalgorithm-diff-perl
libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan8 libbinutils libbct1-0 libctf-nobfd0 libctf0 libdpkg-perl
libfakeroot libfile-fcntllock-perl libgcc-13-dev libgprofng0 libhwscan0 libitm1 liblsan0 libquadmath0 libstdc++13
libstdc++13-dev libubsan1 lto-disabled-list make
Suggested packages:
binutils-doc gprofng-gui debian-keyring g++-multilib g++-13-multilib gcc-13-doc gcc-multilib autoconf automake
libtool flex bison gcc-doc gcc-13-multilib gcc-13-locales gdb-x86-64-linux-gnu git bzr libstdc++13-doc make-doc
The following NEW packages will be installed:
binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev fakeroot g++ g++-13
g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu
libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan8 libbinutils libbct1-0 libctf-nobfd0
libctf0 libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-13-dev libgprofng0 libhwscan0 libitm1 liblsan0
libquadmath0 libstdc++13-dev libubsan1 lto-disabled-list make
0 upgraded, 37 newly installed, 0 to remove and 0 not upgraded.
Need to get 15.2 MB/55.1 MB of archives.
After this operation, 194 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 cdrom://Ubuntu 24.04.3 LTS _Noble Numbat - Release amd64 (20250805.1) noble/main amd64 binutils-common amd64 2.42
-4ubuntu2.5 [240 kB]
Get:2 cdrom://Ubuntu 24.04.3 LTS _Noble Numbat - Release amd64 (20250805.1) noble/main amd64 libstdc++13 amd64 13.2.0-23ubuntu2
```

```
Feb 16 07:05
ubuntu@ubuntu:~
Setting up build-essential (12.10ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
ubuntu@ubuntu:~$ gcc exec_example.c
ubuntu@ubuntu:~$ ./a.out
Child Process
Desktop Documents Downloads Music Pictures Public Templates Videos a.out exec_example.c snap
Parent Process
ubuntu@ubuntu:~$ firefox &
[1] 6522
ubuntu@ubuntu:~$ Gtk-Message: 07:05:03.875: Not loading module "atk-bridge": The functionality is provided by GTK native
ly. Please try to not load it.
[6522, Main Thread] WARNING: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-modul
e.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/
firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201
(firefox_firefox:6522): Gtk-WARNING **: 07:05:04.038: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:05:04.038: Failed to load module "canberra-gtk-module"
[6522, Main Thread] WARNING: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-modul
e.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/
firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201
(firefox_firefox:6522): Gtk-WARNING **: 07:05:04.066: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:05:04.066: Failed to load module "canberra-gtk-module"
ubuntu@ubuntu:~$
```

d) Start a browser and check the process ID of the browser using Top command
firefox &
top

```

Feb 16 07:17
ubuntu@ubuntu:~
bash: kill: (8504) - No such process
ubuntu@ubuntu:~$ nice -n 10 firefox
Gtk-Message: 07:14:39.502: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to
not load it.
[9666, Main Thread] WARNING: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-modul
e.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/
firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201
(firefox_firefox:9666): Gtk-WARNING **: 07:14:39.604: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:14:39.604: Failed to load module "canberra-gtk-module"
[9666, Main Thread] WARNING: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-modul
e.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/
firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201
(firefox_firefox:9666): Gtk-WARNING **: 07:14:39.614: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:14:39.615: Failed to load module "canberra-gtk-module"
ubuntu@ubuntu:~$ top
top - 07:17:10 up 50 min, 1 user, load average: 0.18, 0.80, 1.13
Tasks: 209 total, 1 running, 208 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.4 us, 0.7 sy, 0.0 ni, 98.1 id, 0.0 wa, 0.0 hi, 0.7 si, 0.0 st
MiB Mem : 5926.9 total, 276.0 free, 2369.5 used, 3945.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3557.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
  866 ubuntu    20   0  11.1g 488396 213700 S   2.0   8.0   0:21.88 firefox
  2548 ubuntu   20   0 3499720 412852 156376 S   1.0   6.8   2:56.52 gnome-shell

```

```

Feb 16 07:17
ubuntu@ubuntu:~
top - 07:17:42 up 50 min, 1 user, load average: 0.10, 0.71, 1.08
Tasks: 209 total, 1 running, 208 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 1.1 sy, 0.0 ni, 98.2 id, 0.0 wa, 0.0 hi, 0.7 si, 0.0 st
MiB Mem : 5926.9 total, 276.0 free, 2369.5 used, 3945.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3557.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
  748 ubuntu    20   0 3499720 412852 156376 S   1.3   6.8   2:56.82 gnome-shell
  30 ubuntu    20   0 351956 131636 84440 S   1.0   2.2   1:45.69 Xorg
 2663 ubuntu   20   0 325308 13236  6844 S   0.3   0.2   0:02.36 ibus-daemon
 3179 ubuntu   20   0 354232 29804 18932 S   0.3   0.5   0:04.01 ibus-extension-
 3933 ubuntu   20   0 786372 55708 43296 S   0.3   0.9   0:06.43 gnome-terminal-
    1 root      20   0 23372  14748  9756 S   0.0   0.2   0:03.51 systemd
    2 root      20   0 0 0 0 S   0.0   0.0   0:00.00 kthreadd
    3 root      20   0 0 0 0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-netns
   13 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   15 root      20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   16 root      20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   17 root      20   0 0 0 0 S   0.0   0.0   0:12.09 ksoftirqd/0
   18 root      20   0 0 0 0 I   0.0   0.0   0:01.75 rcu_preempt
   19 root      20   0 0 0 0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
   20 root      20   0 0 0 0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
   21 root      rt   0 0 0 0 S   0.0   0.0   0:00.00 migration/0
   22 root      20   0 0 0 0 S   0.0   0.0   0:00.00 idle_inject/0
   23 root      20   0 0 0 0 S   0.0   0.0   0:00.00 cpuhp/0
   24 root      20   0 0 0 0 S   0.0   0.0   0:00.00 kdevtmpfs

```

e) Kill the browser process using “kill” command
kill PID

```

Feb 16 07:09
ubuntu@ubuntu:~

(Firefox_Firefox:6522): Gtk-WARNING **: 07:05:04.038: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:05:04.038: Failed to load module "canberra-gtk-module"
[6522, Main Thread] WARNING: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-modul
e.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/
firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(Firefox_Firefox:6522): Gtk-WARNING **: 07:05:04.066: GTK+ module /snap/firefox/6565/gnome-platform/usr/lib/gtk-2.0/modu
les/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 07:05:04.066: Failed to load module "canberra-gtk-module"

ubuntu@ubuntu:~$ libEGL warning: DRI2: failed to authenticate

ubuntu@ubuntu:~$ kill PID
bash: kill: PID: arguments must be process or job IDs
ubuntu@ubuntu:~$ kill -9 PID
bash: kill: PID: arguments must be process or job IDs
ubuntu@ubuntu:~$ ^C
ubuntu@ubuntu:~$ ^C
ubuntu@ubuntu:~$ kill PID 6522
bash: kill: PID: arguments must be process or job IDs
ubuntu@ubuntu:~$ Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
[1]+  Terminated                  firefox

```

f) Change the priority of any user process by using “nice” command
 nice -n 10 firefox
 top
 renice 5 PID

```

Feb 16 07:20
ubuntu@ubuntu:~

7500 root      20   0   0   0   0 I   0.3   0.0   0:02.09 kworker/0:0-events
1 root        20   0 23372 14748 9756 S   0.0   0.2   0:03.54 systemd
2 root        20   0   0   0   0 S   0.0   0.0   0:00.00 kthreadd
3 root        20   0   0   0   0 S   0.0   0.0   0:00.00 pool_workqueue_release
4 root        0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
5 root        0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
6 root        0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
7 root        0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-slab_flushwq
8 root        0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-netns
13 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
14 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
15 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
16 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
17 root       0 -20   0   0   0 S   0.0   0.0   0:12.20 ksoftirqd/0
18 root       0 -20   0   0   0 I   0.0   0.0   0:01.76 rcu_preempt
19 root       0 -20   0   0   0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
20 root       0 -20   0   0   0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
21 root       rt   0   0   0   0 S   0.0   0.0   0:00.00 migration/0
22 root       -51  0   0   0   0 S   0.0   0.0   0:00.00 idle_inject/0

ubuntu@ubuntu:~$ renice 5 PID
renice: bad process ID value: PID
ubuntu@ubuntu:~$ ^C
ubuntu@ubuntu:~$ renice 5 PID 7500
renice: bad process ID value: PID
renice: failed to set priority for 7500 (process ID): Operation not permitted
ubuntu@ubuntu:~$ renice 5 7500
renice: failed to set priority for 7500 (process ID): Operation not permitted
ubuntu@ubuntu:~$ renice 10 7500
renice: failed to set priority for 7500 (process ID): Operation not permitted
ubuntu@ubuntu:~$ sudo renice -5 7500
7500 (process ID) old priority 0, new priority -5
ubuntu@ubuntu:~$

```

2. Write a C program using the fork() system call that generates this sequence in the child process. The starting number will be provided from the user. For example, if 8 is passed as a parameter on the command line, the child process will output 8, 4, 2, 1. Because the parent and child processes have their own copies of the data, it will be necessary for the child to output the sequence. Parent invoke the wait() call to wait for the child process to complete before exiting the program. Ensure that a positive integer is passed on the command line.

```
-----  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main(int argc, char *argv[]) {  
  
    if (argc != 2) {  
        printf("Usage: %s <positive integer>\n", argv[0]);  
        return 1;  
    }  
  
    int n = atoi(argv[1]);  
  
    if (n <= 0) {  
        printf("Please enter positive integer\n");  
        return 1;  
    }  
  
    pid_t pid = fork();  
  
    if (pid == 0) {  
        while (n > 0) {  
            printf("%d ", n);  
            n = n / 2;  
        }  
        printf("\n");  
    }  
    else {  
        wait(NULL);  
        printf("Parent exiting...\n");  
    }  
}
```

[illegible]

What is the output of the following program?

```
main()
{
    int a = 10;
    if(fork() == 0)
        a++;
    printf("%d\n",a);
}
```

The following C program is executed on a Unix / Linux system:

```
#include <unistd.h>
int main() {
    int i;
    for (i = 0; i < 10; i++)
        if (i % 2 == 0) fork();
    return 0;
}
```

The total number of child process created is _____.

Q1

Output

10

11

Reason:

- fork() creates child
- Parent → fork() returns >0 → prints 10
- Child → fork() returns 0 → a++ → prints 11
- Separate memory copy

Q2

Even numbers: 0,2,4,6,8 → 5 times fork()

Total processes = $2^5 = 32$

Child processes = 31

4. What happens if parent is killed (status of child), what happens when child is killed, what happens when terminal is killed (). Try these in different sequences - parent, terminal; child terminal; terminal parent etc.

Case 1: Parent killed

Child becomes **Orphan Process**

Adopted by init (PID 1)

Case 2: Child killed

Parent continues normally

Parent may get SIGCHLD

Case 3: Terminal killed

All foreground processes receive SIGHUP

Processes terminate unless running in background or using nohup

5. Process Variables and System Calls

a) Create a child process using fork() and explore the effects of:

exec()
wait()
yield()
sleep()

```
-----  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>
```

```
int main() {  
    fork();  
    sleep(5);  
    return 0;  
}
```

```
-----
```

Effects:

System Call Effect

exec()	Replace process
--------	-----------------

wait()	Parent waits
sleep()	Suspends
yield()	Gives CPU to others

b) Verify whether local, global, and static variables are shared between parent and child processes.

```
-----  
#include <stdio.h>  
#include <unistd.h>  
  
int global = 10;  
  
int main() {  
    int local = 20;  
    static int stat = 30;  
  
    pid_t pid = fork();  
  
    if (pid == 0) {  
        global++;  
        local++;  
        stat++;  
        printf("Child: %d %d %d\n", global, local, stat);  
    }  
    else {  
        sleep(1);  
        printf("Parent: %d %d %d\n", global, local, stat);  
    }  
  
    return 0;  
}
```

c) Verify the parent of a parent process.

```
-----  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {
```

[illegible]